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Surveying—A Glimpse

INTRODUCTION

Surveying is defined as the art of determination of relative position of points on above or below the surface of earth by means of linear and angular measurement, distance, elevation and direction. The relative attitude of points is ascertained by levelling which is another branch of surveying. The maps of the different points of an area are drawn in plan or forms a vertical section with different scales which are essential for different engineering project like roads, highways, railroad, water supply, irrigation, transmission lines, bridges, buildings, dams and reservoirs. It is required for location of different geological structures, showing the geographical boundaries of district, states and countries.

Surveying further may be defined as the technique and science of accurately determining the terrestrial or three-dimensional space position of points and the distances and angles between them. These points are usually, but not exclusively, associated with positions on the surface of the earth, and are often used to establish land maps and boundaries for ownership or governmental purposes. In order to accomplish their objective, surveyors use elements of geometry, engineering, trigonometry, mathematics, physics, and law. An alternative definition, as per the American Congress on Surveying and Mapping (ACSM) is "the science and art of making all essential measurements to determine the relative position of points and/or physical and cultural details above, on, or beneath the surface of the earth, and to depict them in a usable form, or to establish the position of points and/or details".

Furthermore, as alluded above, a particular type of surveying known as land surveying (also per ACSM) is the detailed study or inspection by gathering information through observations, measurements in the field; questionnaires; or research of legal instruments, and data analysis in the support of planning, designing, and establishing of property boundaries. It involves the re-establishment of cadastral surveys and land boundaries based on documents of record and historical evidence, as well as certifying surveys (as required by statute or local ordinance) of subdivision plats/maps, registered land surveys, judicial surveys, and space delineation. Land surveying can include associated services such as mapping and related data accumulation, construction layout surveys, precision measurements of length, angle, elevation, area, and volume, as well as horizontal and vertical control surveys. The analysis and utilization of land survey data

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and its requirement in the planning and execution of nearly every form of construction are very much essential. The most familiar modern uses of surveying are in the fields of transport, building and construction, communications, mapping, and defining legal boundaries for land ownership.

The techniques of conventional surveying and mapping have been revolutionised with the advent of variety of sophisticated instruments in the modern system of surveying and mapping. Such modern instruments and terminology associated with modern surveying are Electronic Distance Measurements (EDM) instruments, Laser (light amplification by stimulated emission of radiation), theodolite, electronic field book, Inertial Positioning System (IPS), Global Positioning System (GPS), Digital Terrain Models (DTMs), automated photographic system, Geographic Information System (GIS), Doppler positioning system and so on.

1.1 HISTORICAL BACKGROUND OF SURVEYING

Surveying has been an essential element in the development of the environment since 5000 years ago. In ancient Egypt, when the Nile river overflowed its banks and washed out farm boundaries, they were reconstructed through the application of simple geometry. Perfect squareness and north-south orientation of the Great Pyramid of Giza built in 2700 BC, affirmed the Egyptians' command of surveying. The Egyptian land register goes back to 3000 BC. Under the Romans, land surveyors were established as a profession, and they established the basic measurements under which the empire was divided, such as a tax register of conquered lands in 300 AD. In England, The Domes Day by William, the Conqueror, in 1086, covered all England, contained names of the land owners, area, land quality, and specific information of the area's content and habitants. Continental Europe's Cadastre was created in 1808 founded by Napoleon Bonaparte. It contained numbers of the parcels of land, land usage, names, etc., and value of the land. Some 100 million parcels of land, triangle survey, measurable survey, map scale: 1:2500 and 1:1250 spread fast around Europe, but faced problems especially in Mediterranean countries, Balkan, and eastern Europe due to cadastre upkeep costs and troubles. A cadastre loses its value if register and maps are not constantly updated. Large-scale surveys are a necessary pre-requisite to map-making. In the late 1780s, a team from the Ordinance Surveys of Great Britain, originally under General William Roy began the principal triangulation of Britain using the specially built Ramsden theodolite.

The Romans developed improved methods of surveying during Roman period. They developed measuring wheel called **odometers** to measure distances. The main development of surveying took place in 19th century after the invention of telescope. More excellent instruments like magnetic compass, levelling instruments and theodolites were developed for making measurement quite quickly and efficiently.

Further development of methods and surveying instruments took place in the 20th century. Precise levelling instruments and modern optical theodolites were invented. Electronic Distance Measurement (EDM) instruments were introduced for accurate measurement of distances. The use of computer, new method of photogrammetric survey, the use of lasers, satellites, applications of GIS, GPS, DTM, remote sensing has brought the subject of surveying to a new and very sophisticated level of study.

1.2 DIVISIONS OF SURVEYING

The shape of earth is an approximate oblate ellipsoid of its revolution. The lengths polar and equatorial axes are 12,713.168 km and 12756.413 km, respectively. The difference is only about 43 km as computed by Clark in 1866. The polar axis is shorter than the equator axis. When the distance of measurement is small, practically there is no difference between the curved distances and corresponding straight line distances, and curvature of the earth is neglected. But in case of bigger area long distances cannot be neglected and earth curvature is considered. Hence, the surface of the earth has the curvature and this is the main factor on which the following two divisions of surveying is based:

1.2.1 Plane Surveying

In plane surveying the curvature of the earth is not taken into account and the surface is treated to be plane. Normally, in small surface area, error due to curvature is significantly much smaller. It can be applied quite safely up to an area of 250 km^2 . Plane surveying is even allowed to conduct to an area upto about 1000 km^2 . Most of the day to day surveys are based on plane surveying. It may be mentioned that the difference between an arc of 18.5 km on the surface of the earth and corresponding chord distance is only 10 mm. Similarly, sum of the angles of spherical triangle having an area of 200 km^2 on the earth surface and that of the corresponding angles of plane triangle is only 1 second, and therefore, for small areas, surveying conducted is mainly plane surveying.

1.2.2 Geodetic Surveying

Geodetic surveying is that type of surveying where the curvature of earth is considered. It is essential if the area of survey is quite large. Actual curvature of the earth is taken into account in this type of survey. The lines on the earth surface are curved; and triangles are spherical. It, therefore, invites spherical trigonometry. Science of geodesy is required in this type of survey. The main object of geodetic surveying is to determine the precise location of widely spaced on the surface of the earth.

Figure 1.1(a) shows three points A , B and C on the mean surface of the earth. The lines $DABE$, $HBCJ$ and $GACF$ are the arcs of the great circles passing through the centre of the earth. The intersection of a plane passing through the centre of the earth forms the great circle. In Figure 1.1(a), the triangle ABC shown by shaded line is the spherical triangle. In Figure 1.1(b) the enlarged view of the spherical triangle ABC of Figure 1.1(a) is depicted. A plane triangle ABC is formed by the dotted lines. The angles a , b and c are the angles of the plane triangle and angles a° , b° and c° are the corresponding angles of the spherical triangle. These angles a° , b° and c° are greater than the angles a , b and c of the plane triangles. The distances AB , BC and CA of spherical triangles in geodetic survey are determined by using spherical trigonometry and in plane triangle by using plane trigonometry.

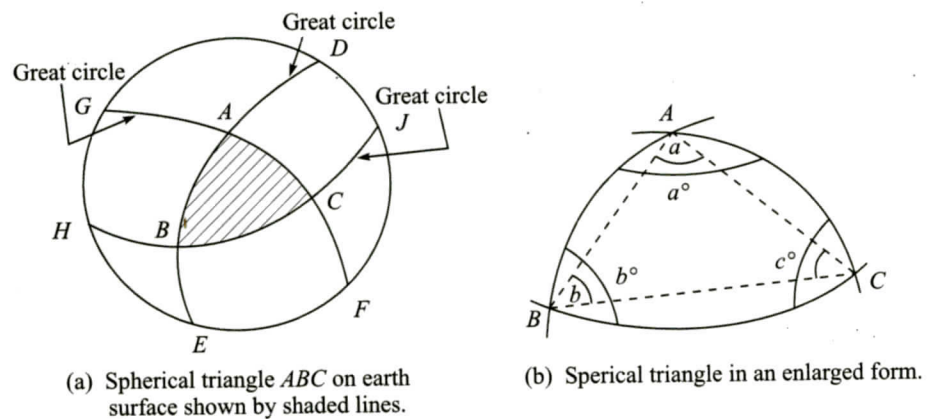


Figure 1.1 The spherical triangle.

Thus, above discussion of these two types of surveying may be concluded as:

✓ (a) **Plane surveying:**

- (i) It is used for relatively small arcs.
- (ii) A curved line on the surface of earth is taken as a straight line.
- (iii) The directions of plumb lines at various points are assumed to be parallel.
- (iv) The spherical angles are considered as plane angles.
- (v) The standard of accuracy is lower than that of geodetic surveying.

✓ (b) **Geodetic surveying:**

- (i) It is used for large areas.
- (ii) It is used for establishing precise control points.
- (iii) The surface of the earth is considered in its exact shape with curvature.
- (iv) The direction of plumb lines at various points are different.
- (v) The accuracy is very high. The instruments used are precise.

✓ **1.3 CLASSIFICATION OF SURVEYING**

The classification of surveying may be based on four dimensions, they are: the function and purpose of surveying; instruments used; method employed; and the place of work.

1.3.1 Classification based on Function and Purpose of Surveying

- (i) Land survey is for determination of areas or boundaries. It may be also called property survey boundary survey or cadastral survey.
- (ii) City survey is for urban planning. It is required for layout of roads, street, sewers, water supply, and layout of buildings in urban areas.
- (iii) Topographic survey is for determination of region such as hills, rivers, lakes, streams, forest, man made features like roads, villages, buildings, canals, and so on.

- (iv) Route surveys are those types which include planning, design, execution of highways, railways, pipe lines canal etc.
- (v) Engineering survey is adopted for engineering purposes. It is done for collection of requisite data for planning design and execution of engineering project like dams, reservoirs tunnels, spillways, roads, bridges, power house, transmission lines, etc. Engineering survey has further stages like reconnaissance preliminary and final location surveys.
- (vi) Military survey is conducted for military purposes for determining points of strategic importance.
- (vii) Mine survey is used for exploring mineral wealth deposited in earth's structure and to guide tunnelling and other operations associated with it.
- (viii) Satellite survey is conducted to establish inter continental inter datum and inter islands geodetic ties all over the world by making observations on artificial satellites.
- (ix) Geological survey is used to determine and explore the components of earth's crust. Geological survey is further conducted for the purpose of recording the geologically significant features of the area under investigation. In the past, in remote areas, there was often no base topographic map available, so the geologist also needed to be a competent surveyor to produce a map of the terrain, on which the geological information could then be draped. More recently, satellite imagery or aerial photography is used as a base, where no published map exists. Such a survey may also be highly specialised - for instance focusing primarily on hydrogeological, geochemical or geomagnetic themes. (Do not confuse this term with Geological survey, typically a government (national, regional or local) body, charged with maintaining and improving the record of the geology of the area in which it operates
- (x) Astronomical surveys are made for determination of attitude, longitude, azimuths, local time, etc., of heavenly bodies such as sun and other fixed star.
- (xi) Control survey is to establish horizontal and vertical positions of widely spaced control points using the principles of geodetic survey.
- (xii) Archaeological survey is done to unearth the relics of antiquity. This survey is used also to accurately assess the relationship of archaeological sites in a landscape or to accurately record finds on an archaeological site.
- (xiii) Gravity survey is carried out to measure the force of gravity at various points.
- (xiv) Foundation survey done to collect the positional data on a foundation that has been poured and is cured.
- (xv) Mortgage survey or physical survey is a simple survey that generally determines land boundaries and building locations. Mortgage surveys are required by title companies and lending institutions when they provide financing to show that there are no structures encroaching on the property and that the position of structures is generally within zoning and building code requirements. Some jurisdictions allow mortgage surveys to be done to a lesser standard, however, most US state minimum standards require the same standard of care for mortgage surveys as any other survey. The resulting higher price for mortgage surveys has led some lending institutions to accept mortgage inspections not signed or sealed by a surveyor.

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- (xvi) Bathymetric survey is a survey carried out to map the seabed profile.
- (xvii) Deformation survey is a survey to determine if a structure or object is changing shape or moving. The three-dimensional positions of specific points on an object are determined; a period of time is allowed to pass. These positions are then re-measured and calculated, and a comparison between the two sets of positions is made.
- (xviii) Soil survey or soil mapping is the process of determining the soil types or other properties of the soil cover over a landscape, and mapping them for others to understand and use.
- (xix) Wetlands delineation and location survey is a survey that is complete when construction work is to be done on or near a site containing defined wetlands. Depending on the local, state, or federal regulations wetlands are usually classified as areas that are completely inundated with water more than two weeks during the growing season

1.3.2 Classification based on Instruments Used

- Chain - Class*
- (i) Chain survey is the simplest type in which linear measurement are made by the chain and the tape. Angular measurements are not required here.
 - (ii) Compass survey is a survey in which angles are to be measured by a magnetic compass and linear measurement by a chain or a tape.
 - (iii) Plane table survey is a survey in which a map of a plot of land is to be prepared in the field in a drawing tapes on the plane table.
 - (iv) Levelling survey is used to determine the elevations of different points of a plot with the help of an instrument called level.
 - (v) Chain and compass survey is used to plot of a land with the help of chain and compass.
 - (vi) Theodolite survey is primarily used in traversing and triangulation for providing controls. The instrument theodolite is used to measure vertical and horizontal angles for plotting curves on roads railways gate.
 - (vii) Tacheometric survey is done with special type of instrument known as tacheometer to determine horizontal and vertical distances.
 - (viii) Photogrammetry or photographic survey is made with the help of photographs.
 - (ix) EDM survey is done by Electronic Distance Measurement (EDM) instrument method. It has been developed quite recently. Instruments are for high waves and radio wave measurement.

✓ 1.3.3 Method Employed

- (i) Triangulation survey is the process of measuring the angles of a chain or network of triangles formed by stations marked on the surface of earth.
- (ii) Traverse survey is that type of survey in which number of survey lines forms the framework and the directions and the lengths of survey lines are measured with the help of angle or direction measuring instruments.

✓ 1.3.4 Place of Work

- (i) Land survey is done to measure objects on earth's surface.

- (ii) Hydrographical or marine survey is done to measure objects under water.
- (iii) Aerial survey is done via aeroplane in air.

1.4 PRINCIPLES OF SURVEYING

The basic principles of surveying have changed over ages, but the tools used by surveyors have evolved tremendously. Engineering, especially Civil Engineering depends too much on surveyors. The surveyors should have a thorough knowledge of algebra, basic calculus, geometry and trigonometry. In addition to the above, surveyors should be able to use all delicate types of instruments with precision and accuracy. The methods of plane surveying are based on the two fundamental principles, as follows:

1.4.1 Locating a Point from Two Other Known Points

The relative positions of points in survey works can easily be located from two known fixed points by only one of the methods described in Figure 1.2(a), (b), (c), (d) and (e).

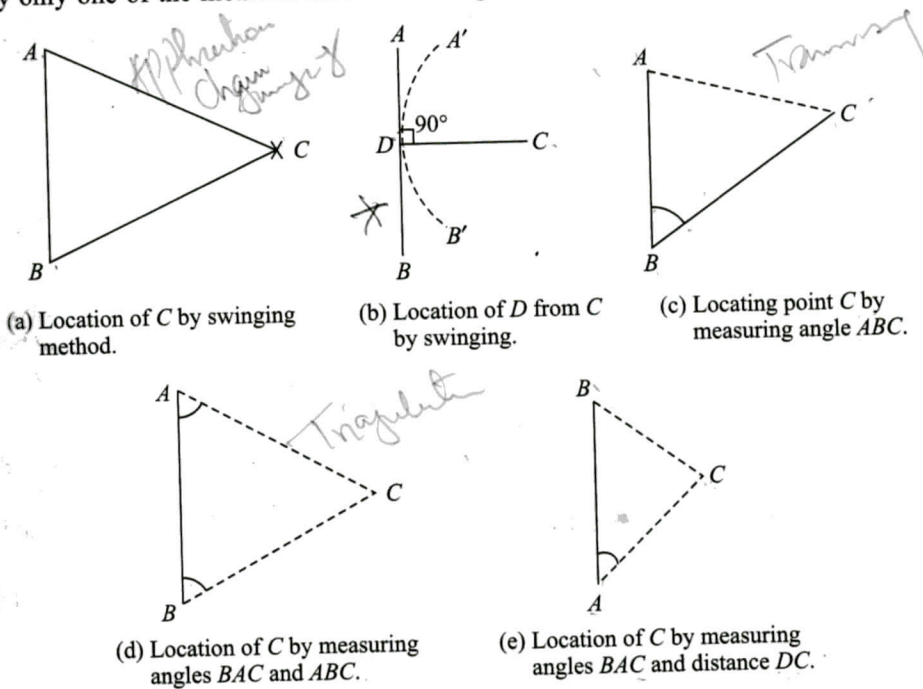


Figure 1.2 Methods of locating a point from two other known points.

- (i) In Figure 1.2(a) the location of points A and B on the ground are known. The locations of point C can be located easily by swinging arc method by measuring AC and BC is commonly used in chain survey.
- (ii) In Figure 1.2(b) a perpendicular CD at point D on AB from C can be drawn by a tape taking minimum distance of CD to AB by swinging the tape. The tape is kept

on swinging along the path $A'DB'$ shown by a dotted line and point D , i.e. CD is the minimum distance of C to the chain line. Thus, CD is perpendicular to AB at D from C . This can be plotted by set squares.

- (iii) In Figure 1.2(c) the distance AC and angle CBA can be measured. Using a protractor or trigonometrical equation point C is plotted. It is used in traverse survey.
- (iv) In Figure 1.2(d) the angle BAC and ABC are measured by an angle measuring instrument. Knowing the distance AB , C is located either by protractor or by solution of trigonometrical equation of triangle. It is used in triangulation survey. Thus, the points can be plotted in drawing sheet to a suitable scale
- (v) In Figure 1.2(e), angle BAC and distance BC are measured and point C is plotted by protractor or trigonometrically or by swinging about the point C . This method is of minor utility and is used in exceptional cases.

1.4.2 Working from the Whole to the Part

In this procedure, surveyor should first establish a sufficient number of points with high degree of precision in and around the area to be surveyed. These are called **primary control point**. The gaps are then filled with secondary control points at closer interval. Further, gaps are then filled up by tertiary control points even at closer intervals. For the survey of ordinary nature, the tertiary control points are used to fix details on the ground. The aim of working in this way is to prevent accumulation of error.

1.5 BASIC MEASUREMENTS IN SURVEYING

The reference line in surveying is the direction of gravity which is required in all measurements. This direction of gravity line is the plumb line, i.e. the line of freely suspended plumb bob which is taken as a vertical line. The plane perpendicular to this gravity line is called **horizontal line**. Surveying basically consists of the following measurements:

1. Horizontal distances: The distance along a horizontal plane is the horizontal distance. If a distance is measured in a slope, it is converted to its horizontal equivalent.
2. Vertical distances: Vertical distance is measured along the direction of gravity. It is essential to find the difference of levels between points.
3. Horizontal angle: Horizontal angle is the angle between two lines in a horizontal plane.
4. Vertical angle: It is the angle between two lines in the same vertical plane.
5. Vertical plane: A plane containing a vertical line is a vertical plane.
6. Level surface: It is the equipotential surface of the earth's gravitational field. It is actually a curve surface and every element of which is normal to the plumb line. A body of still water is the best example of level surface.
7. Level line: A line lying on a level surface is a level line. Thus, it is a curved line normal to the plumb at all points.
8. Horizontal plane: It is a plane which is tangential to the level surface at a particular point.

9. Horizontal line: A line tangent to the level surface is a horizontal line.
10. Vertical line: It is a line perpendicular to the horizontal plane.
11. Zenith angle: An angle between two lines in a vertical where one of the lines directed towards the zenith is known as **zenith angle**.
12. Elevation: The vertical distance of a point from an assumed datum or mean sea level (MSL) is known as an **elevation**.
13. Contour: It is an imaginary line of equal or constant elevation on the surface of the ground.
14. Grade or gradient: The rate of ascent or descent or the slope of the line is called **grade or gradient**.
15. Latitude: If the y-axis in a Cartesian co-ordinate is in north-south direction, y co-ordinate of a point is called **latitude**.
16. Departure: If the x-axis in a Cartesian co-ordinates is in east-west direction, x co-ordinate of a point is called **departure**.

1.6 INSTRUMENTS USED FOR DIFFERENT MEASUREMENTS

Different instruments are used for different measurements as follows:

- (i) Instruments used for horizontal distances are tapes, chains, tacheometers and EDM.
- (ii) Instruments used for vertical distances are levelling instruments, tacheometers, and so on.
- (iii) Instruments used for horizontal angles include magnetic compass, theodolites, sextants, etc.
- (iv) Instruments used for vertical angles are theodolites, clinometers and sextants.

1.7 BASIC METHODS USED IN SURVEYING

In surveying, all the methods involve in measuring angles and distances are for both horizontal and vertical planes. The locations of points required to measure are determined by the applications of various trigonometrical and geometrical processes. In surveying, these processes are known by distinctive names as follows:

1.7.1 Methods for Fixing Position in the Horizontal Plane

Three methods are used for fixing the position. They are intersection, angle and distance and resection.

Intersection

Intersection can be done by intersection of angles (Figure 1.3) and by intersection by distances (Figure 1.4). In the intersection of angles, the points *A* and *B* can be located and its distance can be measured. The position of point *C* can be obtained by intersection relative to *A* and *B*. The angle *CAB* and *CBA* can be measured in the field and position of *C* is obtained by plotting the